

Paper 2

1. Evaluate $\int_0^2 \frac{3}{7x+2} dx$.

2. Find $\int \left(\sin 5x - \frac{3}{e^{2x}} \right) dx$.

3. Use three applications of the trapezoidal rule to estimate the value of $\int_1^4 \sqrt{x-1} \, dx$. Give your answer correct to 4 significant figures.

4. Solve $\sin x = -\frac{1}{2}$, where $0 \leq x \leq 2\pi$. Giving the answer in exact form in radians

5. Find $\int 10^{5x-1} dx$.

6. A group of medical researchers conducted an investigation to study the impact of different dosages of a new drug on the duration of relief from an allergic reaction.

An experiment was carried out using a random sample of seven patients. The data collected is recorded in the table.

<i>Dosage (mL)</i>	3	4	4	5	7	7	9
<i>Duration of relief (hours)</i>	5	9	12	9	18	22	22

a) Find the equation of the least-squares regression line. Give your answer in the form $y = Ax + B$, where A & B are correct to 2 decimal places.

b) By finding Pearson's correlation coefficient, describe the strength and direction of the relationship between the dosage and the duration of relief.

- c) Using the equation found in part (a), estimate the hours of relief with a dosage of 4.3 mL. Give your answer correct to the nearest hour.

7. Evaluate in simplest form $\int_0^{\frac{\pi}{2}} \sin 2x \, dx$.

8. Differentiate the following with respects to x .

a. $\frac{3x + 2}{x + 4}$

b. $x(1 + 2x)^4$

9. A discrete random variable X has the probability distribution table shown below.

$X = x$	5	10	15	20
$P(x)$	0.1	0.25	n	0.35

a) Find the value of n .

b) Find the expected value $E(X)$.

10. Differentiate with respect to x .

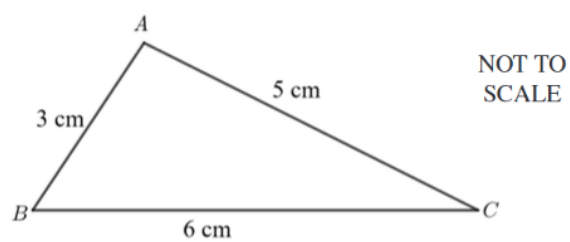
a) $y = (3x^2 - 7)^6$.

b) $y = \frac{2x - 1}{x + 8}$.

11. Find the equation of the tangent to the curve $y = x^3 + \ln x + 5$ at the point $(1, 6)$.

12. Solve the equation $\sqrt{3} + 2\cos\theta = 0$ in the domain $0 \leq \theta \leq 2\pi$

13. ABC is a triangle in which $AB = 3$ cm, $BC = 6$ cm and $CA = 5$ cm as shown in the diagram below.



Use the Cosine rule to calculate $\angle BAC$ to the nearest degree

14. Consider the curve $y = x^3 - 12x^2 + 36x$.

a) Find the x and y intercepts.

b) Find the location and nature of any stationary points.

c) Find the point of inflection.